PIYUSH KHANDELWAL

Senior Frontend Engineer | React · TypeScript · Frontend Architecture · AI Agents · Performance
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PROFESSIONAL SUMMARY

Senior Frontend Engineer with 7 years building production web applications across fintech, enterprise SaaS, and generative-AI products in high-growth teams. Deep expertise in React, Next.js, TypeScript, frontend architecture, micro-frontends, design systems, state management, and performance engineering. Currently a core engineer on Unveilix, an AI-native conversational analytics platform, where I own the AI experience layer over a RAG-grounded LLM agent that turns natural-language business questions into governed SQL and real-time agent workflows. Experienced owning frontend systems end to end—from architecture and product/design collaboration through testing, CI/CD, release safety, and production monitoring. Trusted for system design and pragmatic technical trade-offs, with 100+ PR reviews annually, mentoring experience, and a track record of improving Core Web Vitals and automated test coverage.

TECHNICAL SKILLS

Frontend: React 19, Next.js (15, App Router), TypeScript, JavaScript ES6+, AngularJS, React Native, HTML5/CSS3, Tailwind CSS, SCSS/Sass, Redux Toolkit, Zustand, TanStack Query/Table, Storybook, ECharts, data visualization

Frontend Architecture: Micro-frontends & Module Federation, design systems & component libraries, monorepo architecture, state management patterns, SSR/SSG rendering strategy, code-splitting & bundle optimization, Core Web Vitals (LCP/INP/CLS), Accessibility, Turborepo, Vite

Testing & Quality: Jest, Chai, Mocha, React Testing Library, Cypress, Playwright, Datadog, Sentry, Rollbar, ESLint, Prettier, Husky

Generative AI & LLM: RAG (Retrieval-Augmented Generation), LLM-powered text-to-SQL, AI agents & agentic workflows, prompt engineering, streaming LLM UX (SSE), multi-provider LLM integration (Gemini, OpenAI, Claude), AI-assisted development & coding agents: Claude Code, Cursor, GitHub Copilot

PROFESSIONAL EXPERIENCE

Carnera Technologies Pvt Ltd - Senior Software Developer Pune | Feb 2024 – Present

- Own the frontend architecture: Next.js 15 App Router structure, Zustand state stores, a reusable chart and component library with shared design tokens, and a Jest + React Testing Library suite enforced through a Husky/ESLint/Prettier pre-commit pipeline - keeping a 10+ engineer team shipping to consistent UI standards.
- Engineered the streaming agent interface on a custom POST-based SSE client (the browser's native EventSource is GET-only), consuming 15+ typed agent events—including ReasoningStep, ContextTrace, SchemaDiscovery, SQLGeneration, Validation, QueryExecution, and Summary—into a live "thinking" timeline that delivers sub-second first-token feedback on multi-minute agent runs.
- Built the RAG transparency layer surfacing exactly which tables, columns and business documents grounded each answer, plus metric-conflict resolutions giving analysts citation-level auditability over AI-generated SQL.
- Designed the schema-intelligence and business-context console—AI-generated table/column descriptions with human review, metric/glossary approval, and document upload—curating the corpus that the retrieval layer queries and materially reducing hallucinated joins on ambiguous schemas.
- Authored a deterministic chart-routing engine that reconciles the model's visualization hint against data-shape rules and reports "AI suggested X → switched to Y", preventing bad LLM picks from reaching users while keeping model intent visible; locked the behavior in with golden-file tests.
- Integrated per-thread multi-provider model selection across Gemini, OpenAI, and Claude with admin-gated token-usage metering for workspace-level LLM cost control.
- Shipped drag-and-drop dashboards, KPI cards, drill-down workflows, and scheduled email digests with headless-Chromium PDF rendering.
- Leveraged TanStack Query for server-state management, caching, request deduplication, and background data synchronization across the analytics application, reducing redundant API requests and keeping data-heavy views responsive.
- Act as the go-to engineer for technical guidance and architectural decisions on a 10+ engineer product team—conducting 100+ code reviews annually across frontend and backend, unblocking teammates on production incidents, and setting frontend conventions the team builds against.

- Drive pragmatic trade-off calls that avoid premature over-engineering: replaced an unbounded LLM-driven visualization path with a deterministic rules engine reconciled against model hints - cutting failure modes and inference cost without building a full ML feedback loop - and consistently scope technical-debt paydown against delivery timelines rather than rewriting wholesale.
- Architected and orchestrated a micro-frontend ecosystem within a shared monorepo, decoupling team dependencies and increasing deployment frequency by 40%.
- Engineered a production Property Analyzer using Claude Code for AI-assisted architecture and implementation, applying TypeScript boundaries, testing, and code review to validate and harden agent-generated output.
- Improved Core Web Vitals (LCP/INP) by 25% through strategic code-splitting, tree-shaking and custom caching layers, directly reducing user bounce rates.
- Elevated automated test coverage from 10% to 90% using Playwright and Jest, establishing a "zero-regression" culture within the engineering team.
- Partner with designers from wireframe to release — reviewing Figma files for technical feasibility, flagging costly interactions before build, and translating designs into pixel-accurate, responsive and accessible components backed by shared design tokens.

Bitpack.me - Software Development Engineer II (Frontend) Bengaluru | May 2023 – Jan 2024

- Directed the micro-frontend migration for a decentralized app store, achieving a 20% boost in runtime performance through optimized module federation.
- Integrated Sentry for frontend error monitoring and production diagnostics, tracking runtime exceptions and user-impacting failures to accelerate incident detection and debugging.
- Implemented Datadog observability with Real User Monitoring (RUM), Core Web Vitals, application performance monitoring, and Sentry error tracking, improving visibility into frontend performance, production errors, and user-impacting regressions.
- Engineered a high-performance design system using Tailwind CSS, reusable UI primitives, shared design tokens, and Storybook documentation—standardizing UX across 4+ product lines and improving frontend delivery speed by 30%.
- Improved UI state and rendering stability using Zustand plus memoization/caching patterns, reducing state-related regressions and improving responsiveness.
- Built and maintained React Native experiences alongside web applications, sharing reusable components and frontend patterns while adapting interactions to mobile platform constraints.
- Built and maintained Node.js/Express services backing the app store - REST endpoints, API aggregation and caching layers - and owned the integration contract between those services and the micro-frontend clients.

West Agile Labs - Software Development Engineer (Full Stack) Hyderabad | Jan 2020 – May 2023

- Migrated a React application to Next.js and added unit tests, increasing code coverage to 80% and improving performance/SEO.
- Built and optimized backend services including Node.js REST APIs, caching and queue-based processing, improving throughput by up to 40% and reducing response times (100ms).
- Designed Node.js/Express service layers end to end - schema and endpoint design, authentication, error handling and integration testing - shipping features from database through API to UI as the sole full-stack owner on multiple client projects.
- Designed and built GraphQL APIs—type schemas, queries, mutations, and resolvers—and consumed them on the frontend with Apollo Client, replacing chatty REST round-trips with single typed queries and reducing client-side data-fetching code and payload size.
- Developed and maintained an Electron desktop client leveraging shared web components, integrating backend APIs and improving packaging/release reliability.
- Built complex map editing workflows (polygon create/edit/copy/delete) using Mapbox/Azure Maps, improving editor efficiency and reducing manual errors.

EDUCATION

B.Tech, Computer Science & Engineering - Medi-Caps University, Indore 2020